

Moein Razavi, Ph.D.

Senior AI/ML Engineer | Generative AI & Agentic Systems | Machine Learning | Google Cloud

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SUMMARY

AI/ML engineer with a Ph.D. and 8+ years of Python experience across industry and applied research, including 3+ years at Ford building and running production AI on Google Cloud. Designs multi-agent LLM systems, retrieval-augmented generation (RAG), text-to-SQL, and document-extraction pipelines that have cut manual work by 30–60%, and builds deep learning forecasting models in PyTorch. Delivers end to end, from ambiguous requirements to tested, deployed services (FastAPI, Docker, Cloud Run, Terraform, CI/CD), and has launched iOS and web products as a founder. Published researcher in machine learning, wearable sensing, and deep neural network training acceleration.

TECHNICAL SKILLS

Generative AI & Agents: Multi-agent systems, Google ADK, LangChain, LangGraph, Model Context Protocol (MCP), RAG, BM25 and vector search (FAISS, ChromaDB), text-to-SQL, tool calling, structured outputs, human-in-the-loop approval, LLM output validation; Gemini, GPT-4o, Claude

Machine Learning: PyTorch, TensorFlow/Keras, scikit-learn, XGBoost, Hugging Face, Optuna; transformers (PatchTST, TimesFM, iTransformer), time-series forecasting, computer vision (Faster R-CNN), Monte Carlo simulation, SHAP, Core ML

Languages: Python, SQL, TypeScript/JavaScript, Swift, C++, Java

Backend & Data: FastAPI, Flask, REST, WebSocket, Server-Sent Events, PySpark, Apache Airflow, BigQuery, Dataform, pandas, PostgreSQL, MongoDB, Supabase, Firebase

Cloud & DevOps: Google Cloud (Vertex AI, Cloud Run, BigQuery, Dataproc, Cloud Storage), AWS (EC2, S3, Lambda), Docker, Kubernetes, Terraform, Tekton, GitHub Actions

Security & Frontend: OAuth 2.0, JWT, SAML/ADFS SSO, role-based access control, PII redaction; React/Next.js, Dash, SwiftUI

PROFESSIONAL EXPERIENCE

Ford Motor Company

Aug 2023 – Present

Senior Data Scientist | Remote

- **Multi-agent analytics assistant:** Cut time spent answering recurring analytics and document questions by **30%** with a LangChain/GPT-4o system in which an intent router delegates to a schema-aware text-to-SQL agent over BigQuery and a BM25 RAG agent over legal documents and vehicle manuals; a checker agent validates answers against domain rules. Launched on Cloud Run behind SAML/ADFS single sign-on.
- **Document-to-data ingestion (HOMELoad):** Reduced manual data preparation by **~50%** with a 5-agent Python/FastAPI service that converts inconsistent third-party utility bills (CSV, Excel, PDF, XML, HTML) into a standard schema using Gemini schema discovery, PII redaction, deterministic validation, bounded self-correcting retries, and idempotent BigQuery MERGE loads; deployed with Docker, Cloud Run, and Terraform.
- **AI code-review tooling:** Reduced manual code-review time by **60%** with a FastAPI service that parses repositories (GitPython, Python AST), streams LLM-assisted complexity reviews over Server-Sent Events, and estimates Vertex AI compute cost.
- **Transformer forecasting for energy-market bids:** Reduced vehicle-availability forecast error by **~14%** versus the previous baseline across **~9,500 vehicles** by building PatchTST-style transformers in PyTorch, benchmarking TimesFM, MOIRAI-2, and iTransformer, and tuning with Optuna on Vertex AI GPUs. Monte Carlo capacity estimates and greedy vehicle selection feed ERCOT vehicle-to-grid market bids.
- **Distributed data pipelines:** Build and maintain EV telemetry pipelines processing **10+ TB** across **15 Airflow DAGs** (PySpark on Dataproc, BigQuery, Dataform); debug production data issues end to end, including time-of-use rate and savings calculations behind smart-charging reports.
- **ML services:** Delivered a utility-rate recommendation service with XGBoost bill predictions, Flask REST APIs on BigQuery, and a SAML-authenticated Dash front end, released on Cloud Run through Tekton CI/CD.

CereVu Medical, Inc.

Jun 2023 – Sep 2024

Software Developer / Research Engineer (Contract) | Remote

- Developed and launched **two SwiftUI iOS apps** (Vitality and Clinical) that capture **9 physiological signals** from wearable sensors over CoreBluetooth BLE, distributed through TestFlight and App Store submissions.
- Re-architected the Clinical app from UIKit to MVVM-Coordinator, adding sensor calibration, data export, role-based workflows (patient, physician, administrator), and OAuth 2.0/JWT authentication.
- Built Python/Flask services with WebSocket streaming and MongoDB feeding a React/Next.js clinician dashboard; deployed Docker services on AWS EC2 with GitHub Actions CI/CD.

Texas A&M University

Sep 2018 – May 2024

Graduate Research Assistant (Machine Learning, Wearable Sensing, Computer Vision) | College Station, TX

- **Wearable stress detection (first author):** Built the modeling and evaluation pipeline for Apple Watch heart-rate and motion data from 50+ participants: participant-grouped cross-validation, bootstrap confidence intervals, permutation tests, fixed false-positive-rate operating points, and SHAP explanations across 6 model families.
- **MERCURY (IEEE HPCA 2023, co-author):** Accelerated deep neural network training with random projection, quantization, and cached computation reuse; reported **1.97x** average training speedup across 12 models at baseline-level accuracy.
- **Computer vision (first author):** Built face-mask and physical-distancing monitoring for construction video; fine-tuned TensorFlow Faster R-CNN detectors on a 3,300-image dataset (1,000 images self-collected and annotated) to **99.8%** mask-classification accuracy, with perspective transforms for distance measurement.
- **OpenSync (open source, first author):** Built a Python platform on Lab Streaming Layer that synchronizes EEG, eye-tracking, physiological sensors, and experiment events; published tests measured RMS recording-time lag below 0.07 ms. Released on PyPI and GitHub.

SELECTED PROJECTS

- **Inspecalytics (Founder & Technical Lead, 2023 – Present):** Scoped requirements with inspection clients and launched a product that cut inspection-report preparation time by **~50%**: SwiftUI iOS app, React/TypeScript web app, and Supabase (PostgreSQL) backend with offline edit conflict resolution, Apple Vision/ARKit photo organization, automated Word/PowerPoint reports, and XCTest/Vitest suites.
- **Business Action Prioritizer (Google ADK, Gemini, MCP):** Built a 4-stage agent workflow that turns business spreadsheets and a scoring rubric into ranked actions. A planning agent proposes the analysis and waits for user approval; an execution agent delegates to a PandasAI child agent; an MCP server delivers reports through signed Cloud Storage URLs; FastAPI/SQLite manage sessions.
- **On-device transformer for Apple Watch (mHELP, 2026):** Deployed a PyTorch-trained transformer through Core ML, reimplementing training-time preprocessing in Swift (300-point heart-rate sequences, 18 engineered features, exported scalars) for probability-based stress alerts.
- **Clink Social (Founder):** Built SwiftUI/MapKit discovery and messaging on Firebase, TypeScript Cloud Functions for feeds and notifications, and QR-based cross-device sign-in with time-limited OAuth/JWT tokens.

EDUCATION

Ph.D., Industrial Engineering | Texas A&M University | GPA 4.0/4.0 Dec 2023

Dissertation: Advancing Stress Detection: Real-Time, Naturalistic, and Personalized AI Approaches

Master of Engineering, Computer Engineering | Texas A&M University | GPA 4.0/4.0 May 2021

B.S., Mechanical Engineering | Amirkabir University of Technology (Tehran Polytechnic) Jun 2018

SELECTED PUBLICATIONS

Razavi, M., et al. Machine Learning, Deep Learning, and Data Preprocessing Techniques for Detecting, Predicting, and Monitoring Stress and Stress-Related Mental Disorders: Scoping Review. *JMIR Mental Health*, 2024.

Janfaza, V., Weston, K., **Razavi, M.**, et al. MERCURY: Accelerating DNN Training by Exploiting Input Similarity. *IEEE HPCA*, 2023.

Razavi, M., et al. An Automatic System to Monitor the Physical Distance and Face Mask Wearing of Construction Workers in COVID-19 Pandemic. *SN Computer Science*, 2022.

Razavi, M., et al. OpenSync: An Open-Source Platform for Synchronizing Multiple Measures in Neuroscience Experiments. *Journal of Neuroscience Methods*, 2022.

Peer reviewer: *IEEE Journal of Biomedical and Health Informatics; Transactions of the Institute of Measurement and Control*.